IQOO HACKATHON · BATTLE 04 · HYDERABAD

AI Interview Coach

Practice interviews the way they actually happen — on camera, on mic, under pressure — with an AI panel that scores you like a real recruiter would.

Team: TRINETRA (Solo Build)

Builder: M. Bharath Kumar

Problem Statement: Smart Education

THE PROBLEM

Mock interviews don't feel like real interviews



Text-only practice

Chatbot Q&A skips the two things recruiters actually judge — how you look and sound under pressure.



Feedback is generic

"Good answer, be more confident" doesn't tell a student what to fix or how much they improved.



Zero accountability

Nothing stops a candidate from opening ChatGPT in another tab mid-"interview."



One-size-fits-all Qs

Same generic questions for every student, ignoring their actual resume and the role's JD.

OUR SOLUTION

One AI panel. Camera, mic, resume and integrity checks — together.

AI Interview Coach runs a full mock interview end-to-end: it asks role-specific questions, watches and listens to how you answer, locks the tab so you can't cheat, and hands back a scored report with exactly what to fix.

5	3	100%	1:1
AI-Judged Score Metrics	Input Modes: Cam · Mic · Text	Proctored, Tab-Locked Sessions	Resume-Matched Questions

New for this hackathon build →

Live video presence scoring • Voice-based answering • Tab-switch proctoring • Resume-aware questions • Progress-tracking dashboard

NEW FEATURE 01

Live Video Interview Presence

The AI interviewer turns on your camera and reads the room the way a human panel would.

Eye-contact %

Tracks gaze vs. camera using on-device face landmarks (MediaPipe).

Expression & emotion read

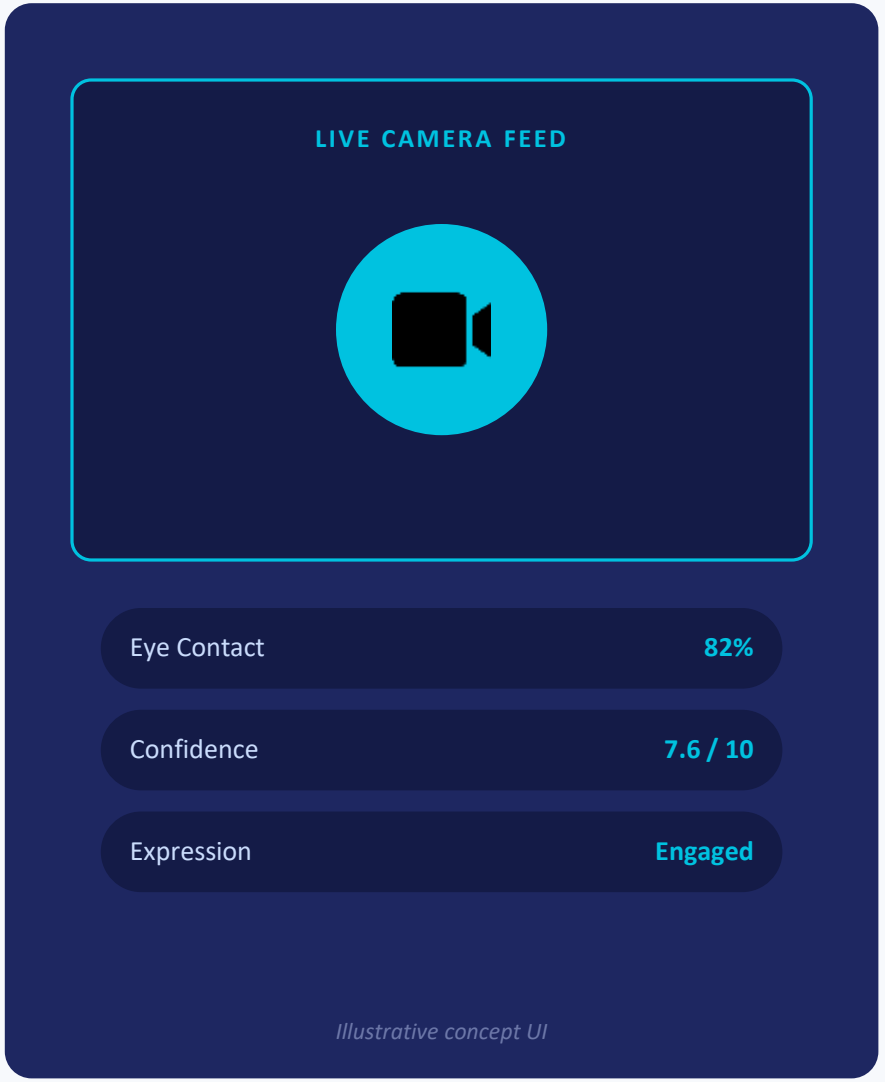
Flags nervousness, low energy, or flat affect frame-by-frame.

Posture & framing

Nudges candidates who slouch, look away, or sit off-frame.

Confidence score

Rolls video signals into one 0–10 body-language score on the report.



NEW FEATURE 02

Voice-First Answering

Tap the mic and answer out loud — exactly how a real interview works. No typing required.



Speech-to-text capture

Browser mic streams straight into Gemini-ready transcripts — no external app switch.



Pace & filler-word tracking

Flags "um/like/so" counts and words-per-minute against an ideal 120–150 WPM band.



Tone & sentiment analysis

Reads confidence, hesitation and clarity from voice, not just word choice.



Communication score

Combines pace, clarity and filler rate into the report's Communication metric.

NEW FEATURE 03

Anti-Cheat Proctoring

A score only means something if it's earned. The session locks down the moment it starts.

Fullscreen lock

Interview only runs in fullscreen — exiting pauses and flags the session.

Tab-switch & blur detection

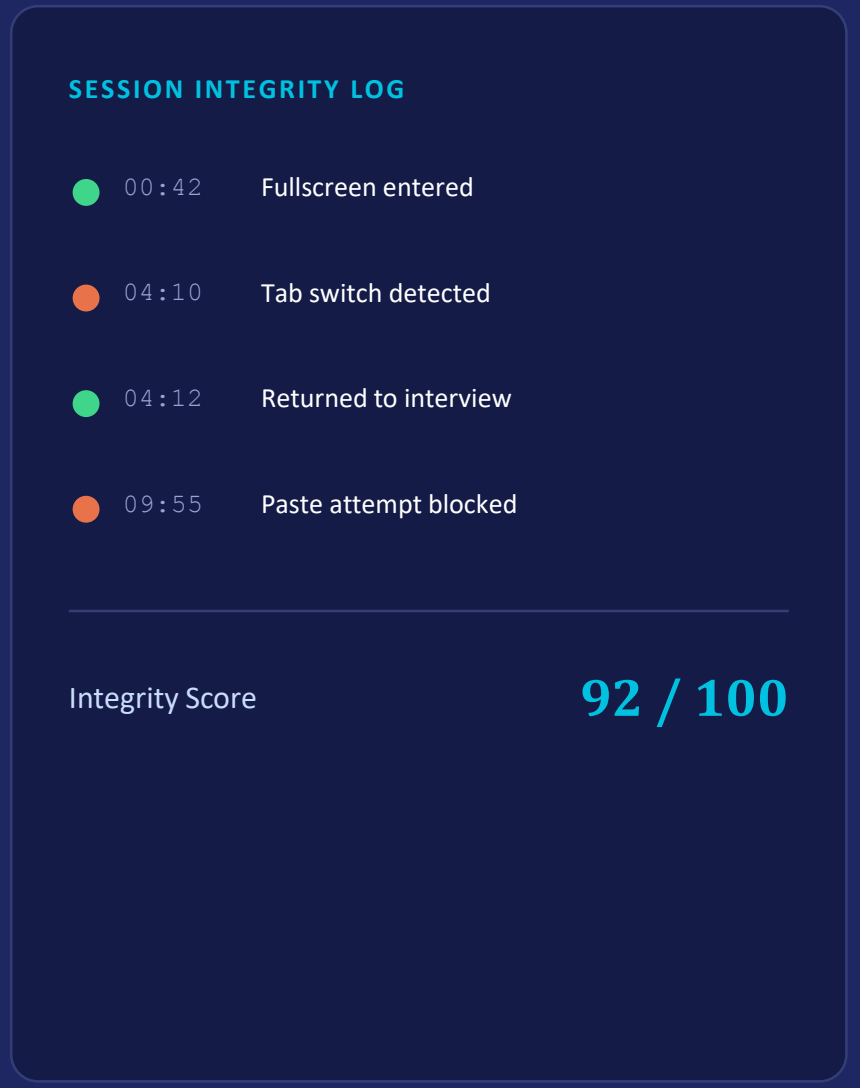
Page Visibility + focus events catch any jump to ChatGPT or another tab.

Copy-paste block

Answer box disables paste, blocking pre-written or AI-generated answers.

Integrity score on report

Every violation is timestamped and shown to the candidate and recruiter.



NEW FEATURE 04

Resume & JD-Aware Question Engine

Upload a resume (PDF/DOCX) and paste the job description — Gemini builds a question set around the actual role and the candidate's real projects.



STEP 1

Upload

Resume (.pdf/.docx) + job description text



STEP 2

Parse

Extract skills, projects & experience with docx/pdf parsers



STEP 3

Generate

Gemini drafts technical, HR & project questions tied to the resume



STEP 4

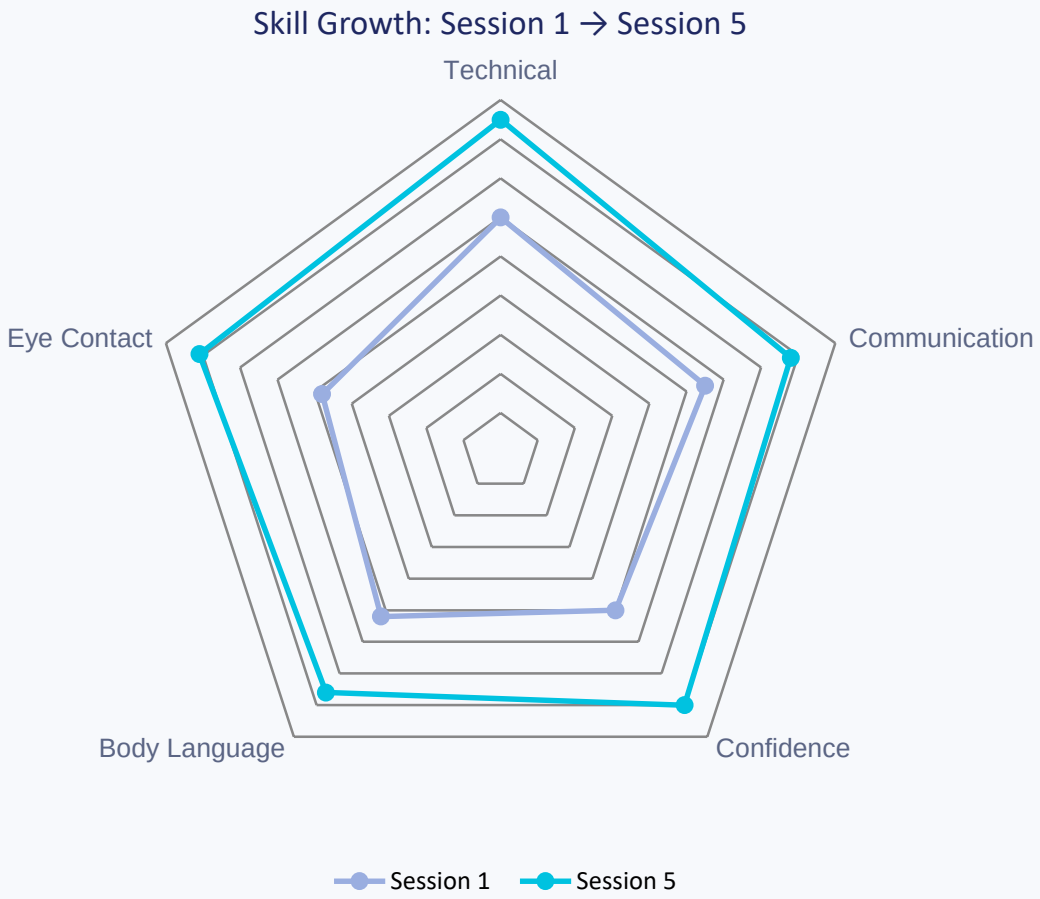
Personalize

Follow-up questions probe specific resume claims in real time

NEW FEATURE 05

Progress-Tracking Analytics Dashboard

Every session's video, voice and answer scores roll into one dashboard — so improvement is visible, not just implied.



Overall Score

8.1 / 10

+2.6 vs. Session 1

STRONGEST AREA

Technical depth

NEEDS WORK

Eye contact & pacing

TECH STACK

Built on a Working Stack, Extended for the Hackathon

ALREADY BUILT

- **Python + Flask**
  App backend & routing
- **SQLite**
  Zero-setup relational store
- **Google Gemini API**
  Question generation & scoring
- **Bootstrap 5 + Chart.js**
  UI & score visualisation

ADDING FOR THE HACKATHON

- **MediaPipe / OpenCV**
  Face, gaze & expression tracking
- **WebRTC + Web Speech API**
  Live camera & mic capture
- **Whisper / Speech-to-Text**
  Voice → transcript pipeline
- **Page Visibility API**
  Tab-switch & focus proctoring

WHY THIS ISN'T JUST A SLIDE

The core product already runs end-to-end

This is a working prototype, not a mockup — 12 pages, 5 database tables, and a live Gemini integration are already built and demoable today.

- OTP-verified registration, login & password reset
- Gemini-generated Technical, HR, Behavioral & Situational questions
- AI evaluation across 4 score metrics with feedback
- Score charts (Chart.js) on a results dashboard
- Admin panel for managing users, interviews & results
- SQLite schema: users, interviews, results, interview_qa, otp_tokens

12	5	4	1
Page templates	Database tables	AI score metrics	Live Gemini integration

EXECUTION PLAN

30-Hour Build Plan, If Shortlisted

Hr 0–6

Video + Voice Core

Wire up WebRTC camera capture, Web Speech API mic input, and stream answers into the existing Gemini pipeline.

Hr 6–14

Proctoring Layer

Add fullscreen lock, tab-visibility & blur listeners, paste-block, and the session integrity log.

Hr 14–22

Resume Engine + Dashboard

Resume/JD upload & parsing, personalized question generation, radar-chart progress dashboard.

Hr 22–30

Polish & Demo

Bug-fix pass, sample data for a clean live demo, and a 3-minute walkthrough script.

PRODUCT ROADMAP

Beyond the Hackathon: The Bigger Vision

The 30-hour build ships the core product. These are the 12 features that turn it into something recruiters and colleges would actually adopt.



Feels Like a Real Interview

- Switchable AI personas — Friendly HR, Strict Tech Lead, Panel
- Live coding round with real-time code review
- Adaptive difficulty that reacts to how you're doing
- Video replay with AI comments at each timestamp



Built for Indian Campus Hiring

- Group Discussion simulator vs. AI participants
- Built-in aptitude & reasoning round
- Company-style modes — Amazon, TCS, Infosys style



Keeps Candidates Coming Back

- Weakness heatmap tracked across sessions
- Streaks & achievement badges
- AI suggests exactly what to study next



Built to Scale

- Recruiter-side dashboard to screen candidates
- Shareable “Interview Ready” certificate for LinkedIn

WHY TRINETRA

A solo builder who ships, not just pitches

Full-stack, solo, already working

Built and deployed the entire Flask + Gemini + SQLite product alone — this pitch is an upgrade plan, not a first draft.

Product thinking, not just code

Every new feature (video, voice, proctoring, resume-matching) maps directly to what recruiters actually evaluate.

Realistic 30-hour scope

The roadmap only adds what's buildable in the hackathon window, on top of a stable existing codebase.

AI Interview Coach — Team TRINETRA

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